

SAMPLE DISCRIMINANTS WHICH MINIMIZE POSTERIOR SQUARED ERROR LOSS*

Peter Enis & Seymour Geisser

State University of New York at Buffalo

1. Introduction

Suppose we have two p-variate nonsingular normal populations where Π_i is $N(\mu_i, \Sigma_i)$ for $i=1,2$. If an observation Z , which prior to its having been observed, could have arisen from Π_i with known prior probability q_i ($0 < q_i < 1$, $q_1 + q_2 = 1$) then the (conditional) probability of coming from Π_i , given $Z = z$, is

$$\Pr[\Pi_i|z] = \frac{q_i f_i(z)}{q_1 f_1(z) + q_2 f_2(z)}$$

where

$$f_i(z) = f(z|\mu_i, \Sigma_i^{-1}) \propto |\Sigma_i^{-1}|^{1/2} \exp(-\frac{1}{2} \text{tr} \Sigma_i^{-1} (z - \mu_i)(z - \mu_i)')$$

A reasonable procedure would be to assign z to that population for which $\Pr[\Pi_i|z]$ was maximum. For convenience, in the normal case we express this by the equivalent rule:

$$(1.1) \quad \begin{aligned} U &= U(\mu_1, \mu_2, \Sigma_1^{-1}, \Sigma_2^{-1}|z) = \log \frac{f_1(z)}{f_2(z)} \geq \log r, \text{ assign } z \text{ to } \Pi_1 \\ U &< \log r, \text{ assign } z \text{ to } \Pi_2 \end{aligned}$$

where $r = \frac{q_2}{q_1}$.

Now, for any classification scheme a commonly accepted viewpoint may be summarized as "the proof of the procedure is in the erring". Indeed, in addition to the reasonableness of (1.1), it is well known (e.g. Anderson, 1958) that the use of U as the discriminant for assigning observations into either Π_1 or Π_2 minimizes the probability of misclassification.

Until now, we have assumed that the parameters, μ_i and Σ_i , of the two populations are known. However, the situation generally encountered in practice is one where not all (if any) of the parameters are known; and, furthermore, the situation most frequently dealt with in the literature is the linear case; i.e., Π_i is $N(\mu_i, \Sigma)$ and μ_1, μ_2 and Σ are all unknown.

*This research was supported in part by NIGMS NIH Grant GM 14031.

Suppose we have a sample of n_i independent observations $x_{i1}, x_{i2}, \dots, x_{in_i}$ from Π_i ($n_1 + n_2 + 1 > p$), $i=1,2$. Let $\bar{x}_i = n_i^{-1} \sum_{j=1}^{n_i} x_{ij}$ and $(n_i - 1)S_i = \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)(x_{ij} - \bar{x}_i)'$. Then, using $\bar{x}_i$ as an estimate for μ_i and the pooled sample covariance matrix $S = (n_1 + n_2 - 2)^{-1} [(n_1 - 1)S_1 + (n_2 - 1)S_2]$ as an estimate for the common covariance matrix Σ , the usual procedure (e.g. Anderson, 1958, and Geisser, 1967) has been to substitute these statistics for the unknown parameters in U forming a sample discriminant, say V, which is used in the same manner as the population discriminant, U, to assign observations. Since for this case

$$(1.2) \quad U = z' \Sigma^{-1} (\mu_1 - \mu_2) - \frac{1}{2} (\mu_1 + \mu_2)' \Sigma^{-1} (\mu_1 - \mu_2)$$

(linear in z), one forms the sample discriminant viz.,

$$(1.3) \quad V = \hat{U} = z' S^{-1} (\bar{x}_1 - \bar{x}_2) - \frac{1}{2} (\bar{x}_1 + \bar{x}_2)' S^{-1} (\bar{x}_1 - \bar{x}_2)$$

and uses (1.1) with U replaced by V. Presumably, one would follow the same procedure in the case of a quadratic discriminant ($\Sigma_1 \neq \Sigma_2$). That is, form V by substituting in U the estimates $\bar{x}_i$ and S_i for μ_i and Σ_i , respectively, and then use V as the discriminator.

Although, as mentioned before, it is well known that the use of U minimizes the probability of misclassification, the use of V cannot be justified in the same manner. However, as Anderson (1958, p. 137) says, "it seems intuitively reasonable" that V "should give good results." Fisher (1936), of course, has also given an independent argument for the use of $z' S^{-1} (\bar{x}_1 - \bar{x}_2)$ (the first term in (1.3) above), which he calls the discriminant function, as the criterion for classifying an observation into one of two populations.

It is clear that the problem can be viewed as one of estimating U in (1.2) and that a good estimator of U would naturally be considered to yield a good discriminator. Pursuing this approach, Geisser (1967) has shown (using the "usual" noninformative prior distribution for μ_1, μ_2 and Σ^{-1}) for the linear discriminant U given by (1.2) that, for fixed z, the posterior mean of U (that value which minimizes the squared error loss function) is

$$E[U|z] = V + \frac{1}{2} p(n_1^{-1} - n_2^{-1})$$

where V is given by (1.3). Thus, from a Bayesian viewpoint, one has an additional justification for the use of V (for this particular linear case) besides the fact that "it seems intuitively reasonable".

In what follows we will assume that we have n_i ($> p$) independent observations $x_{i1}, x_{i2}, \dots, x_{in_i}$ from Π_i which is $N(\mu_i, \Sigma_i)$ and we will show, for a variety of situations besides the linear case with $\Sigma_1 = \Sigma_2 = \Sigma$, μ_1 , and μ_2 all unknown, that (for fixed z) $E[U|z] = V + h$, where V is formed by substituting for the unknown parameters in U their sample estimates and where $h = h(p, n_1, n_2)$ is a function only of the sample sizes, n_i , and the dimensionality, p, of the vector random variables and is such that $h(p, n, n) = 0$.

2. Quadratic Case ($\Sigma_1 \neq \Sigma_2$)

Whereas in the linear case (i.e., $\Sigma_1 = \Sigma_2$) one discriminates between populations on the basis of differences in their means, in the quadratic case discrimination is based on differences in both the mean vectors and covariance matrices.

For this situation where $\Sigma_1 \neq \Sigma_2$, we note that

$$U = \frac{1}{2} \left\{ \log \frac{|\Sigma_1^{-1}|}{|\Sigma_2^{-1}|} + (z - \mu_2)' \Sigma_2^{-1} (z - \mu_2) - (z - \mu_1)' \Sigma_1^{-1} (z - \mu_1) \right\},$$

and we consider the estimation of U for various assumptions on the parameters μ_i and Σ_i^{-1} . In all of the following, we employ the "usual" noninformative priors for the parameters of the multivariate normal distribution.

Case (i). Σ_i^{-1} and μ_i are both unknown.

The joint likelihood for (μ_i, Σ_i^{-1}) is

$$(2.1) \quad L(\mu_i, \Sigma_i^{-1}) \propto |\Sigma_i^{-1}|^{1/2} n_i e^{-\frac{1}{2} \text{tr} \Sigma_i^{-1} [(n_i - 1)S_i + n_i (\bar{x}_i - \mu_i)(\bar{x}_i - \mu_i)']}$$

which upon combining with the assumed joint prior density

$$p(\mu_i, \Sigma_i^{-1}) d\mu_i d\Sigma_i^{-1} \propto |\Sigma_i|^{-\frac{1}{2}(p+1)} d\mu_i d\Sigma_i^{-1},$$

yields

$$P(\mu_i, \Sigma_i^{-1} | \bar{x}_i, S_i) \propto |\Sigma_i^{-1}|^{\frac{1}{2}(n_i - p - 1)} \times \exp \left\{ -\frac{1}{2} \text{tr} \Sigma_i^{-1} [(n_i - 1)S_i + n_i (\mu_i - \bar{x}_i)(\mu_i - \bar{x}_i)'] \right\}$$

as the joint posterior density for μ_i and Σ_i^{-1} . Hence, conditional on Σ_i^{-1} , $\mu_i \sim N(\bar{x}_i, n_i^{-1} \Sigma_i)$. Also, marginally $\Sigma_i^{-1} \sim W[(n_i - 1)S_i^{-1}, n_i - 1]$; i.e., Σ_i^{-1} has the Wishart distribution where if $B \sim W(\Omega, \nu)$ then the density of B is given by

$$f(B | \Omega, \nu) = K(p, \nu) |\Omega^{-1}|^{\frac{1}{2}\nu} |B|^{-\frac{1}{2}(\nu - p - 1)} e^{-\frac{1}{2} \text{tr} B \Omega^{-1}},$$

for B positive definite,

$$= 0, \text{ otherwise,}$$

for Ω positive definite, $\nu \geq p$, and where

$$K(p, \nu) = \{2^{\frac{1}{2}p} \pi^{\frac{1}{2}p} \Gamma(\frac{1}{2}p - 1) \prod_{i=1}^p \Gamma(\frac{1}{2}(\nu + 1 - i))\}^{-1}.$$

Now, the evaluation of $E[U|z]$ may be most conveniently obtained by taking conditional and then unconditional expectations, viz.,

$$E[U|z] = \frac{1}{2} \sum_{i=1}^2 (-1)^i E_{\Sigma_i^{-1}} \{ E_{\mu_i} [(z - \mu_i)' \Sigma_i^{-1} (z - \mu_i) | \Sigma_i^{-1}, z] - \log |\Sigma_i^{-1}| \}.$$

This requires the evaluation of terms such as $E[\log|B|]$ where $B \sim W(\Omega, \nu)$. In fact, as will be shown below,

$$(2.2) \quad E[\log|B|] = K(p, \nu) |\Omega^{-1}|^{\frac{1}{2}\nu} \int |B|^{\frac{1}{2}(\nu - p - 1)} \log|B| e^{-\frac{1}{2}\text{tr } B\Omega^{-1}} dB \\ = \sum_{j=1}^p \Psi[\frac{1}{2}(\nu + 1 - j)] + \log |2\Omega|,$$

where $\Psi(z) = \Gamma'(z)\Gamma^{-1}(z)$ is the psi (digamma) function (see Erdélyi, et al, 1953).

Let the $p \times p$ matrix B be distributed as $W(\Omega, \nu)$. Then the characteristic function of $\log|B|$ is

$$\varphi(t) = E[|B|^{it}] = |2\Omega|^{it} \prod_{j=1}^p \frac{\Gamma[\frac{1}{2}(\nu + 2it + 1 - j)]}{\Gamma[\frac{1}{2}(\nu + 1 - j)]}$$

and

$$\varphi'(t) = i\varphi(t) \left\{ \sum_{j=1}^p \Psi[\frac{1}{2}(\nu + 2it + 1 - j)] + \log |2\Omega| \right\}.$$

Hence we obtain

$$E[\log|B|] = -i\varphi'(0) = \sum_{j=1}^p \Psi[\frac{1}{2}(\nu + 1 - j)] + \log |2\Omega|$$

as was asserted.

Before proceeding with the evaluation of $E[U|z]$, we note that (2.2) is a generalization of formula 1, Table 353, page 496 of the tables of Bierens de Haan (1957) wherein is found

$$\int_0^\infty e^{-qx} x^{p-1} \log x \, dx = q^{-p} \Gamma(p) \{ \Psi(p) - \log q \}.$$

Now, it is well known that if the p -vector X is distributed as $N(\delta, \Omega)$ then $Y = (X - a)' \Omega^{-1} (X - a)$ is distributed as $\chi^2(p, \beta)$ where $\beta = (\delta - a)' \Omega^{-1} (\delta - a)$ and $\chi^2(p, \beta)$ denotes the random variable having the noncentral χ^2 distribution with p degrees of freedom and noncentrality parameter β . Moreover, $E[Y] = E[\chi^2(p, \beta)] = \beta + p$. Also, it is easy to show that if $B \sim W(\Omega, \nu)$ then for any nonnull vector of real constants $a' = [a_1, \dots, a_p]$, $Y = \frac{a' B a}{a' \Omega a} \sim \chi_p^2 = \chi^2(\nu; 0)$. Hence, $E[a' B a] = \nu a' \Omega a$.

Since $\Sigma_i^{-1} \sim W[(n_i - 1)^{-1} S_i^{-1}, n_i - 1]$, we obtain

$$E[\log |\Sigma_i^{-1}|] = \log |S_i^{-1}| + \sum_{j=1}^p \{ \Psi[\frac{1}{2}(n_i - j)] + \log 2(n_i - 1)^{-1} \}.$$

Further, $\mu_i \sim N(\bar{x}_i, n_i^{-1} \Sigma_i)$ conditional on Σ_i^{-1} , thus

$$\begin{aligned} E_{\Sigma_1^{-1}} \{ E_{\mu_1} [(z - \mu_1)' \Sigma_1^{-1} (z - \mu_1) | \Sigma_1^{-1}; z] \} &= E_{\Sigma_1^{-1}} [n_1^{-1} p + (z - \bar{x}_1)' \Sigma_1^{-1} (z - \bar{x}_1)] \\ &= n_1^{-1} p + (z - \bar{x}_1)' \Sigma_1^{-1} (z - \bar{x}_1). \end{aligned}$$

Hence, $E[U|z] = V + h(p, n_1, n_2)$ where

$$V = \frac{1}{2} \left\{ \log \frac{|\Sigma_1^{-1}|}{|\Sigma_2^{-1}|} + (z - \bar{x}_2)' \Sigma_2^{-1} (z - \bar{x}_2) - (z - \bar{x}_1)' \Sigma_1^{-1} (z - \bar{x}_1) \right\}$$

and

$$(2.3) \quad h(p, n_1, n_2) = \frac{1}{2} \sum_{j=1}^2 \sum_{i=1}^p (-1)^i \{ \log(n_i - 1) + n_i^{-1} - \Psi[\frac{1}{2}(n_i - j)] \}.$$

It is clear from (2.3) that $h(p, n, n) = 0$. Hence, when the sample sizes are equal, U is unbiased for V (a posteriori).

Case (ii). Σ_1^{-1} is known, μ_1 is unknown.

Combining the prior density for μ_1 ,

$$(2.4) \quad p(\mu_1) d\mu_1 \propto d\mu_1,$$

with (2.1), the likelihood, we obtain

$$P(\mu_1 | \bar{x}_1, \Sigma_1^{-1}) \propto e^{-\frac{1}{2} n_1 (\mu_1 - \bar{x}_1)' \Sigma_1^{-1} (\mu_1 - \bar{x}_1)}$$

as the posterior density for μ_1 .

Since $\mu_1 \sim N(\bar{x}_1, n_1^{-1} \Sigma_1)$ we obtain

$$E[(z - \mu_1)' \Sigma_1^{-1} (z - \mu_1) | z] = n_1^{-1} p + (z - \bar{x}_1)' \Sigma_1^{-1} (z - \bar{x}_1),$$

which in turn yields $E[U|z] = V + h(p, n_1, n_2)$ where

$$(2.5) \quad V = \frac{1}{2} \left\{ \log \frac{|\Sigma_1^{-1}|}{|\Sigma_2^{-1}|} + (z - \bar{x}_2)' \Sigma_2^{-1} (z - \bar{x}_2) - (z - \bar{x}_1)' \Sigma_1^{-1} (z - \bar{x}_1) \right\}$$

and

$$(2.6) \quad h(p, n_1, n_2) = \frac{1}{2} p (n_2^{-1} - n_1^{-1})$$

is zero for $n_1 = n_2$.

Case (iii). Σ_1^{-1} is unknown, μ_1 is known.

We assume that Σ_1^{-1} has prior density

$$(2.7) \quad p(\Sigma_1^{-1}) d\Sigma_1^{-1} \propto |\Sigma_1|^{1/2(p+1)} d\Sigma_1^{-1}.$$

This leads to the posterior density

$$P(\Sigma_1^{-1} | S_1^*; \mu_1) \propto |\Sigma_1^{-1}|^{1/2(n_1 - p - 1)} e^{-1/2 \operatorname{tr} \Sigma_1^{-1} [n_1 S_1^*]},$$

where, since μ_1 is known, we use

$$S_1^* = n_1^{-1} \sum_{j=1}^{n_1} (x_{1j} - \mu_1)(x_{1j} - \mu_1)'$$

as our estimate for Σ_1 .

Since $\Sigma_1^{-1} \sim W(n_1^{-1} S_1^{*-1}, n_1)$, it is clear that

$$E[\log |\Sigma_1^{-1}|] = \sum_{j=1}^p \Psi[1/2(n_1 + 1 - j)] - \log |1/2 \cdot n_1 S_1^*|$$

and

$$E[(z - \mu_1)' \Sigma_1^{-1} (z - \mu_1) | z] = (z - \mu_1)' S_1^{*-1} (z - \mu_1).$$

Hence,

$$(2.8) \quad V = 1/2 \left\{ \log \frac{|S_1^{*-1}|}{|S_2^{*-1}|} + (z - \mu_2)' S_2^{*-1} (z - \mu_2) - (z - \mu_1)' S_1^{*-1} (z - \mu_1) \right\}$$

and

$$(2.9) \quad h(p, n_1, n_2) = 1/2 \sum_{i=1}^2 \sum_{j=1}^p (-1)^i \{ \log n_i - \Psi[1/2(n_i + 1 - j)] \},$$

where, once again, $h(p, n, n) = 0$.

Case (iv). Σ_1^{-1} is unknown, $\mu_1 = \mu_2 = \mu$ is known.

It is clear that this is just a special version of Case (iii). Hence, V is obtained by putting $\mu_1 = \mu_2 = \mu$ in (2.8). $h(p, n_1, n_2)$, of course, remains unchanged from that as given in (2.9).

In the four cases thus far considered, we have shown that $E[U|z] = V + h(p, n_1, n_2)$; and it was noted that in all cases $h(p, n, n) = 0$. Furthermore, since for large x the function $\Psi(x)$ is asymptotically equal to $\log x + o(1)$, it is clear that $\lim_{n_1 \rightarrow \infty, n_2 \rightarrow \infty} h(p, n_1, n_2) = 0$.

In the case to be considered now, where Σ_1^{-1} is known and $\mu_1 = \mu_2 = \mu$ is unknown, it

will be seen that the bias factor is not only a function of p , n_1 and n_2 but is also a function of the known parameters Σ_1^{-1} and Σ_2^{-1} . Moreover, for equal sample sizes, contrary to the previous results, this bias does not vanish. This turn of events is a consequence of the fact that we are estimating a common mean with samples from populations with differing covariance matrices. Since h reflects the "weighting of the information" from each population and since in this case the weighting is due not only to the number of observations but also to their dispersion, the equalizing of weight contributed by the sizes of the samples is not sufficient to cause this "weight factor" (bias) to vanish.

As a final remark before presenting the above-mentioned case, we note that the use of the use of the term "weight factor" rather than "bias" may have been a better choice of words with which to describe the function h , because of the connotation imputed to the word "bias". As a matter of fact, if one's primary concern is in estimating U with minimum posterior squared error loss, then the appropriate estimate for U would be $V + h$ and not V alone.

Case (v). Σ_1^{-1} is known, $\mu_1 = \mu_2 = \mu$ is unknown.

The joint likelihood of $(\mu, \Sigma_1^{-1}, \Sigma_2^{-1})$ is

$$L(\mu, \Sigma_1^{-1}, \Sigma_2^{-1}) \propto |\Sigma_1^{-1}|^{1/2} n_1 |\Sigma_2^{-1}|^{1/2} n_2 \\ \times \exp \left\{ -\frac{1}{2} \sum_{i=1}^2 \text{tr} \Sigma_i^{-1} [(n_i - 1) S_i + n_i (\mu - \bar{x}_i) (\mu - \bar{x}_i)'] \right\}.$$

Letting the prior density for μ be given by (2.4) we find that, a posteriori, μ is distributed as $N(M, \Omega)$, where

$$M = (n_1 \Sigma_1^{-1} + n_2 \Sigma_2^{-1})^{-1} (n_1 \Sigma_1^{-1} \bar{x}_1 + n_2 \Sigma_2^{-1} \bar{x}_2) \text{ and}$$

$$\Omega = E[(\mu - M)(\mu - M)'] = (n_1 \Sigma_1^{-1} + n_2 \Sigma_2^{-1})^{-1}. \text{ Since}$$

$$E[(\mu - z)' \Sigma_1^{-1} (\mu - z)] = \text{tr} \Sigma_1^{-1} E[(\mu - z)(\mu - z)'] \\ = \text{tr} \Sigma_1^{-1} E[(\mu - M)(\mu - M)' + (\mu - M)(M - z)' + (M - z)(\mu - M)' + (M - z)(M - z)'] \\ = \text{tr} \Sigma_1^{-1} \Omega + (z - M)' \Sigma_1^{-1} (z - M)$$

we obtain

$$V = \frac{1}{2} \left\{ \log \left| \frac{\Sigma_1^{-1}}{\Sigma_2^{-1}} \right| + (z - M)' \Sigma_2^{-1} (z - M) - (z - M)' \Sigma_1^{-1} (z - M) \right\}$$

and

$$h(p, n_1, n_2, \Sigma_1^{-1}, \Sigma_2^{-1}) = \frac{1}{2} \text{tr} (\Sigma_2^{-1} - \Sigma_1^{-1}) \Omega.$$

Equivalently, letting $\lambda_j (> 0)$, $j = 1, \dots, p$, denote the characteristic roots of $\Sigma_1^{-1} \Sigma_2$, we may write for the above

$$h(p, n_1, n_2, \Sigma_1^{-1}, \Sigma_2^{-1}) = \frac{1}{2} \sum_{j=1}^p \frac{1 - \lambda_j}{n_1 \lambda_j + n_2}.$$

For the sake of completeness, we now return to the linear case (which was considered by Geisser (1967) for the case where μ_1 , μ_2 and Σ , the common covariance matrix, are all unknown) and show that, as before, $E[U|z] = V + h$ for different assumptions on μ_1 and Σ .

3. Linear Case ($\Sigma_1 = \Sigma_2 = \Sigma$)

Case (i). Σ^{-1} is unknown, μ_1 is known.

If we assume a prior density for Σ^{-1} as given by (2.7), then we obtain

$$P(\Sigma^{-1} | S^*, \mu_1, \mu_2) \propto |\Sigma^{-1}|^{1/2} (n_1 + n_2 - p - 1) e^{-\frac{1}{2} \text{tr} \Sigma^{-1} [(n_1 + n_2) S^*]}$$

as the posterior density for Σ^{-1} , where (since μ_1 and μ_2 are known)

$$S^* = (n_1 + n_2)^{-1} \sum_{i=1}^2 \sum_{j=1}^{n_i} (x_{ij} - \mu_1) (x_{ij} - \mu_1)'$$

is used as our estimate for the common Σ . Since

$$\Sigma^{-1} \sim W[(n_1 + n_2)^{-1} S^*, n_1 + n_2]$$
 we obtain

$$E[U|z] = \frac{1}{2} [(z - \mu_2)' S^{*-1} (z - \mu_2) - (z - \mu_1)' S^{*-1} (z - \mu_1)] = V.$$

Hence, in this case, h is identically zero; i.e., U is unbiased for V for arbitrary n_1 and n_2 .

Case (ii). Σ^{-1} is known, μ_1 is unknown.

This is the last nonvacuous case remaining. However, this is merely a special version of Case (ii) of the preceding section. Hence V is obtained by setting $\Sigma_1 = \Sigma_2 = \Sigma$ in (2.5), while $h(p, n_1, n_2)$ remains unchanged from that as given by (2.6).

REFERENCES

ANDERSON, T.W. (1958). *An Introduction to Multivariate Statistical Analysis*. Wiley, New York.

- BIERENS DE HAAN, D. (1957). *Nouvelles Tables D'Intégrales Définies*. Hafner, New York.
- ERDÉLYI, A., et al. (1953). *Higher Transcendental Functions, I*. McGraw-Hill, New York.
- FISHER, R.A. (1936). The use of multiple measurements in taxonomic problems. *Ann Eugen.* 7, 179-188.
- GEISSER, S. (1967). Estimation associated with linear discriminants. *Ann. Math. Statist.* 38, 807-817.

Manuscript received 23rd April 1970.